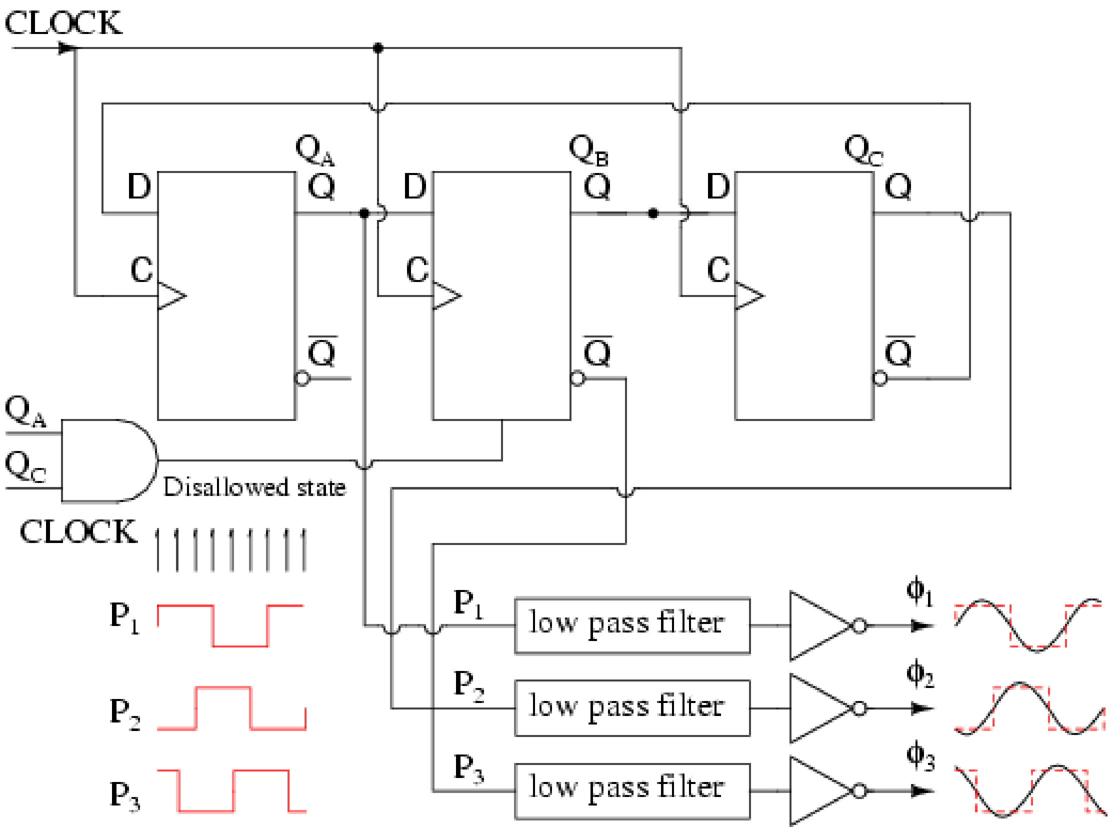
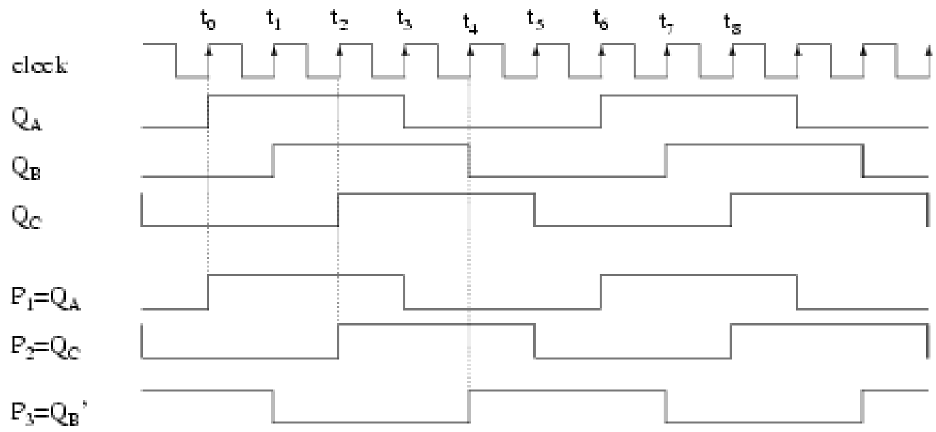
similar capacitor should be applied across the power and ground pins of the '4017.



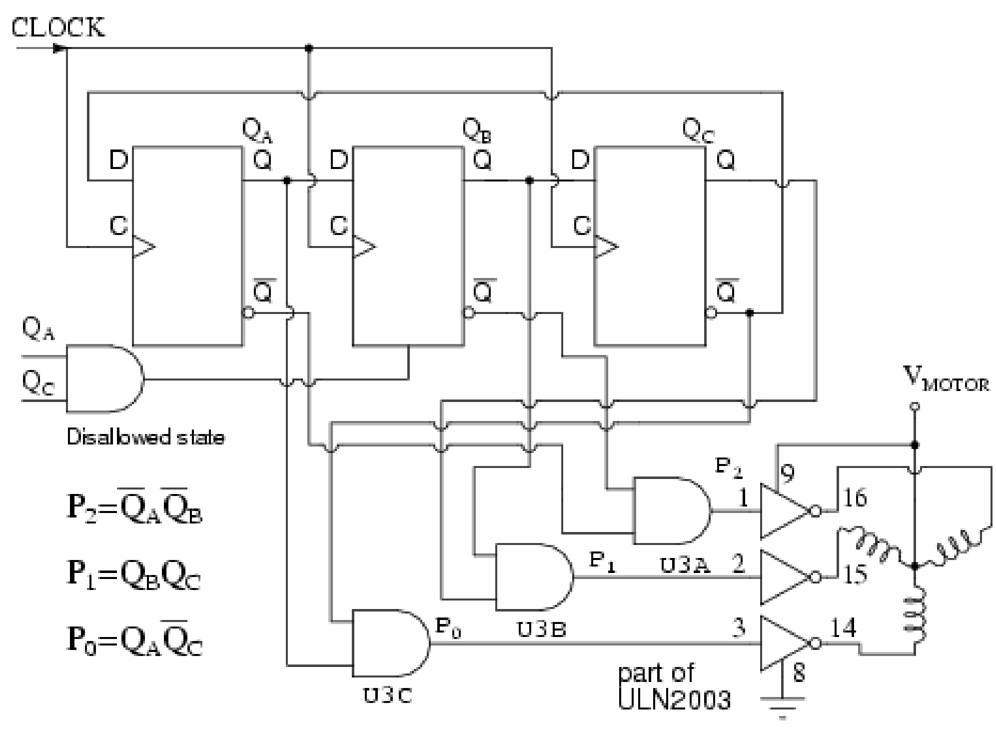

Three phase square/ sine wave generator.

The Johnson counter above generates 3-phase square waves, phased 60$^o$ apart with respect to (**$Q_A$ $Q_B$ $Q_C$**). However, we need 120$^o$ phased waveforms of power applications (see Volume II, AC). Choosing **$P_1$=$Q_A$ $P_2$=$Q_C$ $P_3$=$Q_B$'** yields the 120$^o$ phasing desired. See figure below. If these (**$P_1$ $P_2$ $P_3$**) are low-pass filtered to sine waves and amplified, this could be the beginnings of a 3-phase power supply. For example, do you need to drive a small 3-phase 400 Hz aircraft motor? Then, feed 6x 400Hz to the above circuit **CLOCK**. Note that all these waveforms are 50% duty cycle.




3-stage Johnson counter generates 3-Ø waveform.

The circuit below produces 3-phase nonoverlapping, less than 50% duty cycle, waveforms for driving 3-phase stepper motors.




3-stage (6-state) Johnson counter decoded for 3-ϕ stepper motor.

Above we decode the overlapping outputs $\mathbf{Q_A\ Q_B\ Q_C}$ to non-overlapping outputs $\mathbf{P_0\ P_1\ P_2}$ as shown below. These waveforms drive a 3-phase stepper motor after suitable amplification from the milliamp level to the fractional amp level using the ULN2003 drivers shown above, or the discrete component Darlington pair driver shown in the circuit